NECK POSTURE CORRECTOR

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Introduction

In today's digital world, a large number of people spend most of their time sitting at desks using computers or laptops. This lifestyle has led to a growing issue known as poor posture, especially forward head posture, which often results in neck pain and discomfort. Studies show that neck pain is extremely common among office workers, with nearly 47%–49% of workers experiencing it within a year due to prolonged sitting and screen usage.

This problem is not just theoretical—it is something I personally experience. Even when I try to maintain a correct posture, after some time I unknowingly return to a slouched position. This repeated cycle of correction and relapse made me realize that awareness alone is not enough. There needs to be a practical solution that can help maintain proper posture consistently.

The motivation behind this project comes from my own experience. I often suffer from neck pain due to long hours of using my laptop. Even when I consciously try to sit straight, I eventually fall back into poor posture without noticing.

This made me realize that many people face the same issue. It is not just about knowing the correct posture—it is about maintaining it consistently. That is why I decided to work on this project.

Many individuals, especially students and professionals, suffer from neck pain caused by long hours of sitting in incorrect posture. The main issue is not just poor posture, but the inability to maintain correct posture over time. People often start sitting properly but gradually return to unhealthy positions without realizing it.

Existing solutions like posture correctors exist, but they are often uncomfortable or not effective for long-term habit building. This highlights the need for a better approach that not only corrects posture but also improves awareness.

Objective of project

The goal of this project is to design and develop a neck posture corrector that helps users maintain proper posture while working at a desk. The system should:

- Detect incorrect neck posture
- Provide real-time feedback
- Help users build better posture habits
- Be comfortable and easy to use

Sensors Used

Arduino Uno

Flex Sensor

Motor Vibrator

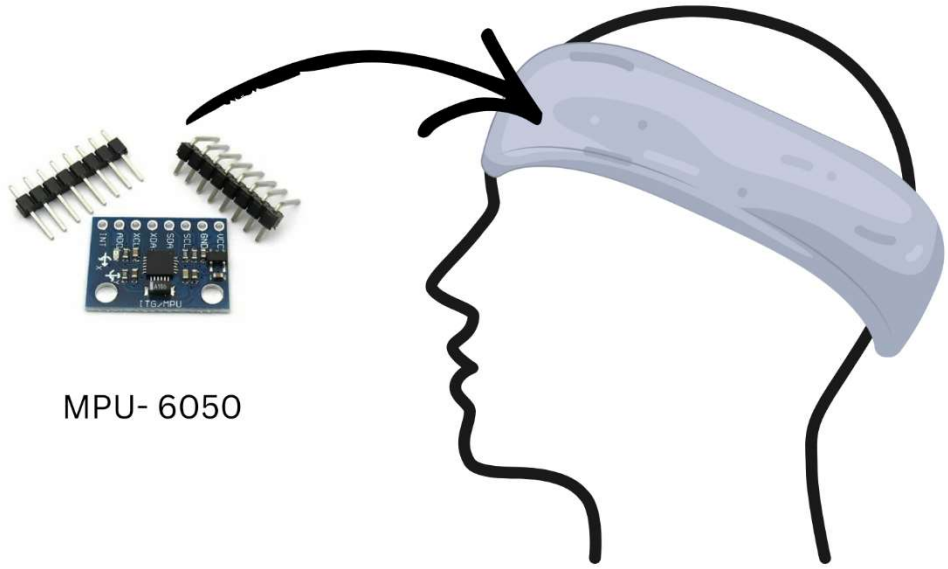
Design Development and key decision

Initial approach

At the beginning, I planned to use the MPU6050 accelerometer. The MPU6050 is a motion-tracking sensor that measures acceleration and rotation across X, Y, and Z axes. My idea was to place it on a headband to detect head movement and posture.

However, I realized a major issue: it detects all movement, not just neck posture. Normal head movements also triggered readings, making it inaccurate for this purpose.

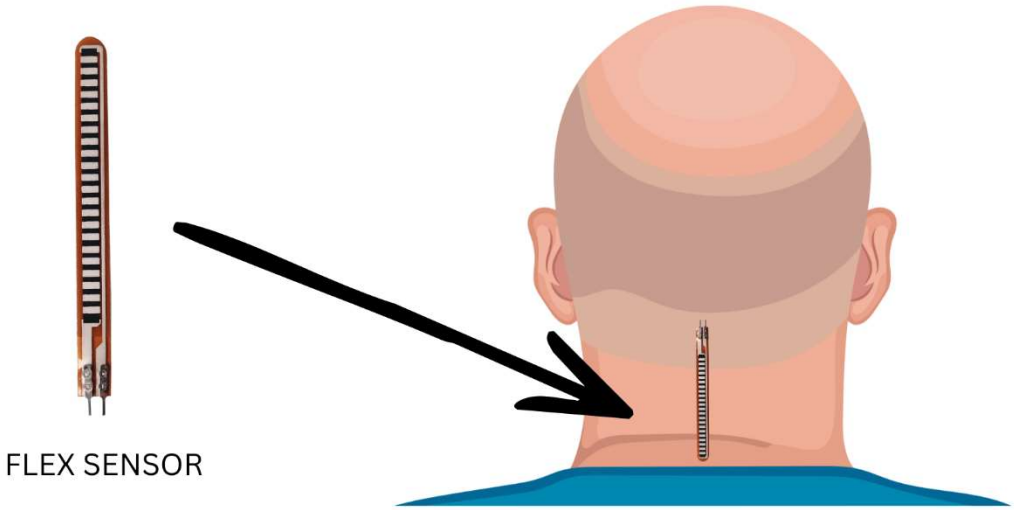
MPU6050 SENSOR WITH INITIAL IDEA SKETCH



Switch to flex sensor

After this, I switched to a flex sensor. A flex sensor changes resistance when bent, making it more suitable for detecting neck bending specifically.

Unlike the accelerometer, it focuses directly on bending rather than general movement, making it more accurate for posture detection.



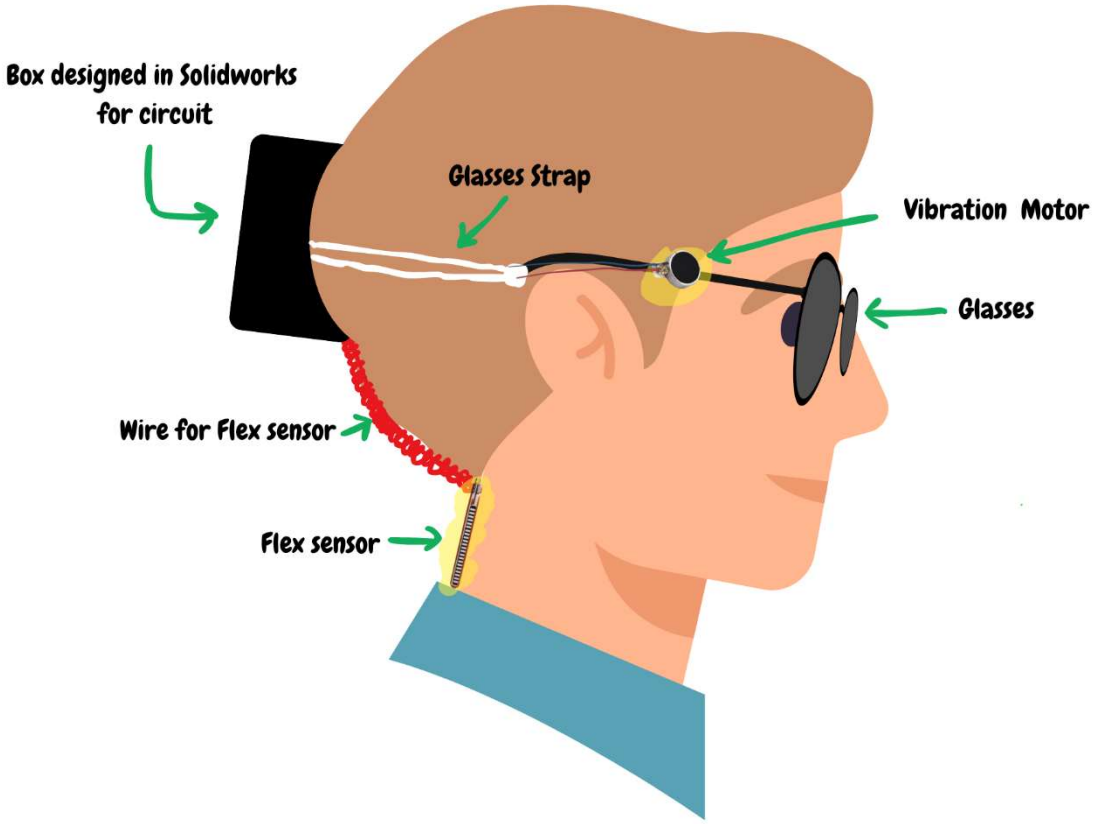
Final Design Concept

After research, I decided to design a system that connects with glasses. Since people already wear glasses while working, this makes the system more practical.

The flex sensor is connected through a wire to a small box designed in SolidWorks, which contains the Arduino and circuit. The sensor extends toward the neck area to measure bending. To inform the user about incorrect posture, a small vibration motor is used. When the flex sensor detects excessive bending, the Arduino activates the vibration motor.

When the user corrects their posture, the vibration stops. This creates a simple feedback loop.

Initially, I considered a piezo buzzer, but it was too loud and could disturb others. A vibration motor is more private and practical.



Step-by-Step Development Process

Step 1: Identifying the Problem

I observed my own posture habits and confirmed that poor posture during long screen time is a real issue.

Step 2: Research

I studied existing posture correctors and found limitations like discomfort and lack of awareness features.

Step 3: Brainstorming Ideas

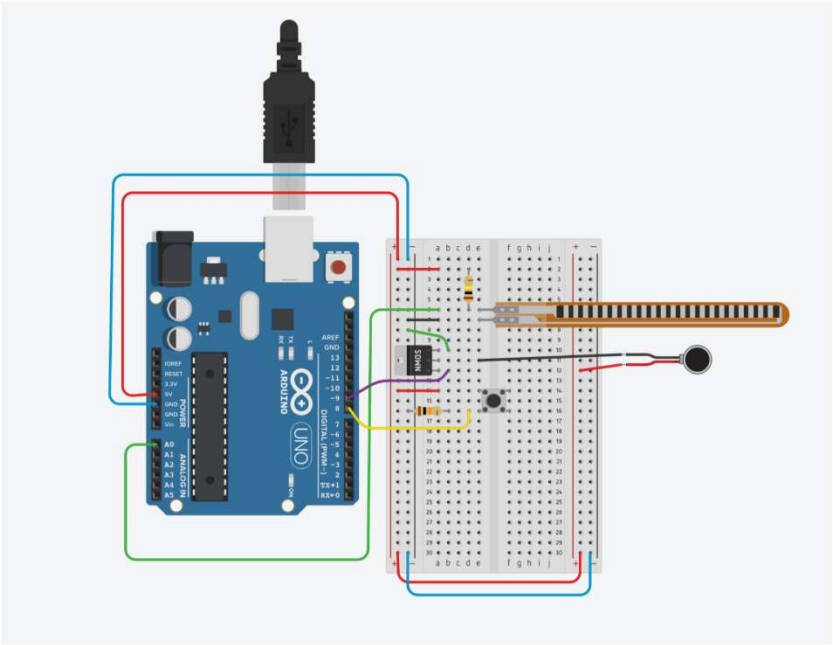
I considered both a physical support system and a smart detection system and chose the smart approach.

Step 4: Designing the System

I planned a sensor-based system that detects neck bending and provides feedback.

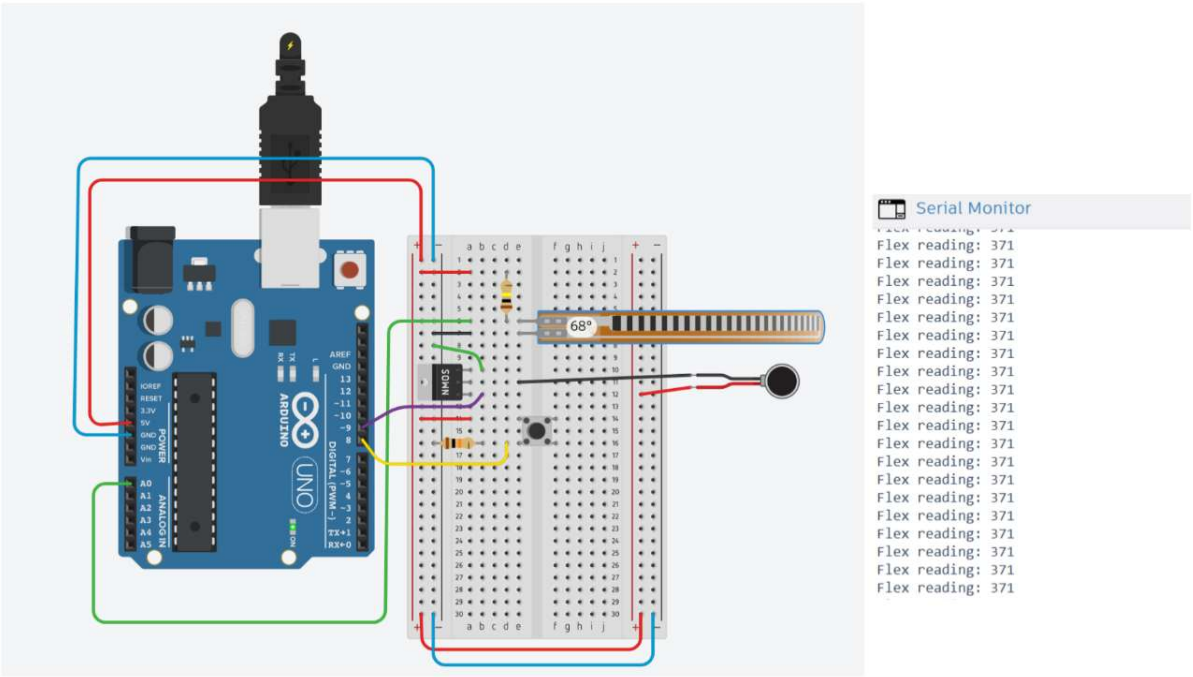
Step 5: Building Prototype

I assembled Arduino, sensor, and vibration motor and tested basic functionality.



Step 6: Testing and Improvements

I tested the system while using a laptop and adjusted sensitivity for better accuracy.



Challenges Faced

One of the major challenges I faced during this project was selecting the right sensor for accurate posture detection. Initially, I chose the MPU6050 accelerometer because it is widely used for motion tracking and seemed like a good fit for detecting head position. However, after analyzing its behavior, I realized that it detects all types of movement, including normal head motions like looking around. This made it difficult to isolate only the incorrect neck posture, leading to inaccurate and unreliable results. This forced me to rethink my approach and switch to a flex sensor, which was more suitable for detecting actual neck bending.

Another challenge was designing a solution that is practical and comfortable for real-world use. At first, I considered placing the sensor on a headband, but this approach did not feel natural or comfortable for long-term use. It took multiple iterations and brainstorming to come up with a more user-friendly design. Eventually, I decided to integrate the system with glasses and position the flex sensor toward the neck area, but reaching this solution required a lot of trial and error.

Calibration of the flex sensor was also a difficult part of the project. I needed to determine a proper threshold value that could accurately differentiate between correct and incorrect posture. If the system was too sensitive, it would trigger alerts even during slight or normal movements, which could annoy the user. On the other hand, if the sensitivity was too low, it would fail to detect poor posture. Finding the right balance required repeated testing and adjustments.

Another important challenge was deciding how to provide feedback to the user. Initially, I thought of using a piezo buzzer, but I quickly realized that it would be too loud and disruptive, especially in shared environments like classrooms or libraries. This led me to switch to a vibration motor, which provides a more subtle and personal form of feedback. While this improved the usability, integrating and testing the vibration system also required additional effort.

Overall, these challenges played a crucial role in improving the final design. Each difficulty forced me to rethink my decisions and refine my approach, ultimately leading to a more practical and effective posture correction system.

Future Improvements

1. Smartphone Connectivity

The system can be connected to a phone to track posture data throughout the day. This will help users see whether their posture is improving over time.

2. Better Wearable Design

The device can be improved to make the flex sensor more comfortable and easy to wrap around the neck area.

Code:

```
const int FLEX_PIN    = A0;
const int MOTOR_PIN   = 9;
const int SWITCH_PIN  = 8;


const int GOOD_POSTURE_VALUE = 200;
const int THRESHOLD          = 250;

const int  MOTOR_SPEED       = 200;
const long SAMPLE_INTERVAL   = 100;


bool systemOn       = false;
bool lastSwitchState = HIGH;
unsigned long lastSample = 0;

void setup() {
  pinMode(MOTOR_PIN,  OUTPUT);
  pinMode(SWITCH_PIN, INPUT_PULLUP);
  analogWrite(MOTOR_PIN, 0);
  Serial.begin(9600);
  Serial.println("Neck Posture Corrector ready.");
  Serial.println("Press the switch to turn the system ON.");
}

void loop() {
  handleSwitch();

  if (systemOn) {
    unsigned long now = millis();
    if (now - lastSample >= SAMPLE_INTERVAL) {
      lastSample = now;
      checkPosture();
    }
  } else {
    analogWrite(MOTOR_PIN, 0);
  }
}
```

```
void handleSwitch() {
  bool currentState = digitalRead(SWITCH_PIN);


  if (lastSwitchState == HIGH && currentState == LOW) {
    delay(20);
    if (digitalRead(SWITCH_PIN) == LOW) {
      systemOn = !systemOn;
      Serial.print("System: ");
      Serial.println(systemOn ? "ON" : "OFF");
    }
  }
  lastSwitchState = currentState;
}

void checkPosture() {
  int flexValue = analogRead(FLEX_PIN);
  Serial.print("Flex reading: ");
  Serial.println(flexValue);

  if (flexValue < THRESHOLD) {
    analogWrite(MOTOR_PIN, MOTOR_SPEED);
    Serial.println("⚠ Bad posture — vibrating!");
  } else {
    analogWrite(MOTOR_PIN, 0);
  }
}
```

Conclusion

This project helped me understand how technology can solve real-life problems in a practical way. Neck pain due to poor posture is a common issue, especially among students and office workers.

Through this project, I learned how to refine ideas, test different approaches, and improve designs based on real-world limitations. Starting from an accelerometer-based idea and evolving into a flex sensor system showed me the importance of adaptability in engineering design.

The final system successfully detects poor posture and provides vibration feedback to help users correct it. Although there is still room for improvement, the project achieves its main goal of increasing posture awareness and encouraging better habits.

In the future, adding mobile connectivity and improving wearability can make this system even more effective.